

2. Matrix Algebra

2.1 Matrix Operations

- **Main Diagonal:** The diagonal entries in matrix $A = [a_{ij}]$ are $a_{11}, a_{22}, \dots, a_{nn}$. They form the main diagonal of A .
- **Diagonal Matrix:** A diagonal matrix is a square $n \times n$ matrix whose non-diagonal entries are zero.

Theorem 1

Let A, B, C be matrices of the same size, and let r and s be scalars.

- $A + B = B + A$
- $(A + B) + C = A + (B + C)$
- $A + 0 = A$
- $r(A + B) = rA + rB$
- $(r + s)A = rA + sA$
- $r(sA) = (rs)A$

Definition (Matrix Multiplication)

If A is an $m \times n$ matrix, and if B is an $n \times p$ matrix with columns $\vec{b}_1, \vec{b}_2, \dots, \vec{b}_p$, then the product AB is the $m \times p$ matrix whose columns are $A\vec{b}_1, A\vec{b}_2, \dots, A\vec{b}_p$. That is:

$$AB = A \begin{bmatrix} \vec{b}_1 & \vec{b}_2 & \dots & \vec{b}_p \end{bmatrix} = \begin{bmatrix} A\vec{b}_1 & A\vec{b}_2 & \dots & A\vec{b}_p \end{bmatrix}$$

Each column of AB is a linear combination of the columns of A using weights from the corresponding column of B .

Row-Column Rule for Computing AB

If the product AB is defined, then the entry in row i and column j of AB is the sum of the products of corresponding entries from row i of A and column j of B .

If $(AB)_{ij}$ denotes the (i, j) -entry in AB , and if A is an $m \times n$ matrix, then:

$$(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{in}b_{nj}$$

- $\text{row}_i(AB) = \text{row}_i(A) \cdot B$
- $\text{column}_j(AB) = A \cdot \text{column}_j(B)$

Theorem 2

Let A be an $m \times n$ matrix, and let B and C have sizes for which the indicated sums and products are defined.

- $A(BC) = (AB)C$
- $A(B + C) = AB + AC$
- $(B + C)A = BA + CA$
- $r(AB) = (rA)B = A(rB)$
- $I_m A = AI_n = A$

Warnings

1. In general, $AB \neq BA$
2. The cancellation law does not hold for matrix multiplication: $AB = AC$ does not imply $B = C$ (in general)
3. If a product AB is the zero matrix, we can't conclude that either $A = 0$ or $B = 0$ (in general)

Definition (Matrix Powers)

If A is an $n \times n$ matrix and if k is a positive integer, then:

$$A^k = \underbrace{A \cdot A \cdot \dots \cdot A}_{k \text{ times}}$$

- $A^{km} = (A^k)^m$
- If $AB = BA$, then:
 - $(AB)^k = A^k B^k$
 - $(A \pm B)^2 = A^2 \pm 2AB + B^2$
 - $(A - B)(A + B) = A^2 - B^2$

Theorem 3 (Transpose Properties)

- a. $(A^T)^T = A$
- b. $(A + B)^T = A^T + B^T$
- c. $(rA)^T = rA^T$
- d. $(AB)^T = B^T A^T$

The transpose of a product of matrices equals the product of their transposes in the reverse order.

2.2 The Inverse of a Matrix

- **Invertible Matrix:** An $n \times n$ matrix A is invertible if there is an $n \times n$ matrix A^{-1} so that $A^{-1}A = I_n$ and $AA^{-1} = I_n$. A^{-1} is uniquely determined by A .
- **Singular Matrix:** A matrix that is not invertible.
- **Nonsingular Matrix:** A matrix that is invertible.

Theorem 4

For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the inverse is:

$$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

- The quantity $ad - bc$ is the determinant of A , denoted $\det A = ad - bc$.
- A 2×2 matrix is invertible if and only if $\det A \neq 0$. If $ad - bc = 0$, then A is not invertible.

Theorem 5

If A is invertible, then the equation $Ax = b$ has the unique solution $x = A^{-1}b$.

Theorem 6

- If A is an invertible matrix, then A^{-1} is invertible and $(A^{-1})^{-1} = A$
- If A and B are invertible $n \times n$ matrices, then so is AB , and $(AB)^{-1} = B^{-1}A^{-1}$
- $(A^T)^{-1} = (A^{-1})^T$
- If λ is a nonzero scalar, then $(\lambda A)^{-1} = \lambda^{-1}A^{-1} = \frac{1}{\lambda}A^{-1}$

The power of a matrix

A is an invertible $n \times n$ matrix and k is a nonnegative integer, then

- $A^{-k} = (A^{-1})^k$
- $(A^\lambda)^\mu = A^{\lambda\mu}$
- $A^0 = I$
- $A^\lambda A^\mu = A^{\lambda+\mu}$

The product of nonsingular (invertible) matrices is invertible, and the inverse is the product of their inverses in the reverse order: $(A_1 A_2 \cdots A_p)^{-1} = A_p^{-1} \cdots A_2^{-1} A_1^{-1}$

Elementary Matrices

- If an elementary row operation is performed on an $m \times n$ matrix A , the resulting matrix can be written as EA , where the $m \times m$ matrix E is created by performing the same row operation on I_m .
- Each elementary matrix E is invertible.
- The inverse of E is the elementary matrix of the same type that transforms E back into I .

Theorem 7

An $n \times n$ matrix A is invertible if and only if A is row equivalent to I_n , and any sequence of elementary row operations that reduces A to I_n also transforms I_n into A^{-1} .

Algorithm for Finding A^{-1}

- Row reduce the augmented matrix $[A \mid I_n]$
- If A is row equivalent to I_n , then $[A \mid I_n]$ row reduces to $[I_n \mid A^{-1}]$
- Otherwise, A is not invertible.

2.3 Characterizations of Invertible Matrix

Theorem 8: The Invertible Matrix Theorem

Let A be a square $n \times n$ matrix. Then the following statements are equivalent (all true or all false):

- A is an invertible matrix
- A is row equivalent to the $n \times n$ identity matrix I_n
- A has n pivot positions
- The equation $Ax = 0$ has only the trivial solution
- The columns of A form a linearly independent set
- The linear transformation $\vec{x} \rightarrow A\vec{x}$ is one-to-one
- The equation $Ax = \vec{b}$ has at least one solution for each $\vec{b}$ in $\mathbb{R}^n$
- The columns of A span $\mathbb{R}^n$
- The linear transformation $\vec{x} \rightarrow A\vec{x}$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$

- j. There is an $n \times n$ matrix C so that $CA = I_n$
 k. There is an $n \times n$ matrix D so that $AD = I_n$
 l. $A^\top$ is an invertible matrix

Square matrices and its inverse

Let A and B be inverse matrices. If $AB = I$, the A and B are both invertible, with $B = A^{-1}$ and $A = B^{-1}$

Theorem 9 (Invertible Linear Transformations)

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear transformation and let A be the standard matrix for T . Then T is invertible if and only if A is an invertible matrix. The linear transformation S given by $S(\vec{x}) = A^{-1}\vec{x}$ is the unique function satisfying:

- $S(T(\vec{x})) = \vec{x}$ for all $\vec{x}$ in $\mathbb{R}^n$
- $T(S(\vec{x})) = \vec{x}$ for all $\vec{x}$ in $\mathbb{R}^n$

Definition

A linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is invertible if there exists a function S such that $S(T(\vec{x})) = \vec{x}$ for all $\vec{x}$ in $\mathbb{R}^n$ and $T(S(\vec{x})) = \vec{x}$ for all $\vec{x}$ in $\mathbb{R}^n$. We call S the inverse of T , and write it as T^{-1} .

2.4 Partitioned Matrices

Example 1: Partitioned Matrix Representation

The matrix

$$A = \left[\begin{array}{ccc|ccc} 3 & 0 & -1 & 5 & 9 & -2 \\ -5 & 2 & 4 & 0 & -3 & 1 \\ -8 & -6 & 3 & 1 & 7 & -4 \end{array} \right]$$

can also be written as a 2×3 **partitioned (or block) matrix**:

$$A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \end{bmatrix}$$

Transpose of a Partitioned Matrix

If A is a partitioned matrix:

$$A = \begin{bmatrix} A_{11} & \dots & A_{1r} \\ \vdots & \ddots & \vdots \\ A_{s1} & \dots & A_{sr} \end{bmatrix}$$

then its transpose A^T is:

$$A^T = \begin{bmatrix} A_{11}^T & \dots & A_{s1}^T \\ \vdots & \ddots & \vdots \\ A_{1r}^T & \dots & A_{sr}^T \end{bmatrix}$$

Theorem 10: Column-Row Expansion of AB

If A is $m \times n$ and B is $n \times p$, then:

$$AB = [\text{col}_1(A) \quad \text{col}_2(A) \quad \dots \quad \text{col}_n(A)] \begin{bmatrix} \text{row}_1(B) \\ \text{row}_2(B) \\ \vdots \\ \text{row}_n(B) \end{bmatrix}$$
$$= \text{col}_1(A)\text{row}_1(B) + \text{col}_2(A)\text{row}_2(B) + \dots + \text{col}_n(A)\text{row}_n(B)$$

Partitioned A (by rows) & B (by columns)

Let $A = (a_{ij})_{m \times s}$ (partitioned by rows):

$$A = \begin{bmatrix} \vec{a}_1^T \\ \vec{a}_2^T \\ \vdots \\ \vec{a}_m^T \end{bmatrix}$$

Let $B = (b_{ij})_{s \times n}$ (partitioned by columns):

$$B = [\vec{b}_1 \quad \vec{b}_2 \quad \dots \quad \vec{b}_n]$$

Then:

$$AB = \begin{bmatrix} \vec{a}_1^T \\ \vec{a}_2^T \\ \vdots \\ \vec{a}_m^T \end{bmatrix} [\vec{b}_1 \quad \vec{b}_2 \quad \dots \quad \vec{b}_n] = \begin{bmatrix} \vec{a}_1^T \vec{b}_1 & \vec{a}_1^T \vec{b}_2 & \dots & \vec{a}_1^T \vec{b}_n \\ \vec{a}_2^T \vec{b}_1 & \vec{a}_2^T \vec{b}_2 & \dots & \vec{a}_2^T \vec{b}_n \\ \vdots & \vdots & \ddots & \vdots \\ \vec{a}_m^T \vec{b}_1 & \vec{a}_m^T \vec{b}_2 & \dots & \vec{a}_m^T \vec{b}_n \end{bmatrix}$$

Block Diagonal Matrix

A **block diagonal matrix** is a partitioned matrix with zero blocks off the main diagonal (of blocks).

Such a matrix is invertible if and only if each block on the diagonal is invertible.

2.5 Matrix Factorizations

Definition of LU Factorization

Assume that A is an $m \times n$ matrix that can be row reduced to echelon form, without row interchanges. Then A can be written in the form $A = LU$, where:

- L is an $m \times m$ **lower triangular matrix** with 1's on the diagonal (called a **unit lower triangular matrix**, and it is invertible),
- U is an $m \times n$ echelon form of A .

This decomposition is called an **LU factorization** of A .

Solving $A\vec{x} = \vec{b}$ via LU Factorization

When $A = LU$, the equation $A\vec{x} = \vec{b}$ can be rewritten as $L(U\vec{x}) = \vec{b}$. Let $\vec{y} = U\vec{x}$, then solve the pair of equations:

$$\begin{cases} L\vec{y} = \vec{b} \\ U\vec{x} = \vec{y} \end{cases}$$

Steps:

1. First solve $L\vec{y} = \vec{b}$ for $\vec{y}$:
Row reduce the augmented matrix $[L \ \vec{b}] \rightarrow [I \ \vec{y}]$
2. Then solve $U\vec{x} = \vec{y}$ for $\vec{x}$:
Row reduce the augmented matrix $[U \ \vec{y}] \rightarrow [I \ \vec{x}]$

Derivation via Elementary Matrices

Suppose A can be reduced to echelon form U using only **row replacements** (add a multiple of one row to another row below it). If there exist unit lower triangular elementary matrices

$E_1, \dots, E_p$ such that:

$$E_p \cdots E_1 A = U$$

then:

$$A = (E_p \cdots E_1)^{-1} U = LU$$

where:

$$L = (E_p \cdots E_1)^{-1} = E_1^{-1} \cdots E_p^{-1}$$

The row operations that reduce A to U will also reduce L to I .

Algorithm for LU Factorization

1. Reduce A to an echelon form U by a sequence of row replacement operations (if possible).
2. Place entries in L such that the same sequence of row operations reduces L to I .

Example 2: LU Factorization of a Matrix

Given matrix:

$$A = \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ -4 & -5 & 3 & -8 & 1 \\ 2 & -5 & -4 & 1 & 8 \\ -6 & 0 & 7 & -3 & 1 \end{bmatrix}$$

Step 1: Reduce A to echelon form U

Row reduce A via row replacements:

$$A \sim A_1 = \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & -9 & -3 & -4 & 10 \\ 0 & 12 & 4 & 12 & -5 \end{bmatrix} \sim A_2 = \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 7 \end{bmatrix} \sim U = \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 5 \end{bmatrix}$$

Step 2: Construct L (unit lower triangular matrix)

From the row operations, L is:

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 1 & -3 & 1 & 0 \\ -3 & 4 & 2 & 1 \end{bmatrix}$$

2.8 Subspaces of $\mathbb{R}^n$

Definition: Subspace

A subspace of $\mathbb{R}^n$ is any set H in $\mathbb{R}^n$ that has three properties:

- The zero vector is in H
 - For each $\vec{u}$ and $\vec{v}$ in H , the sum $\vec{u} + \vec{v}$ is in H (closed under addition)
 - For each $\vec{u}$ in H and each scalar c , the vector $c\vec{u}$ is in H (closed under scalar multiplication)
- The set $\text{Span}\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_p\}$ is called the subspace spanned (or generated) by $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_p$.

Definition: Column Space

The column space of a matrix A , denoted $\text{Col } A$, is the set of all linear combinations of the columns of A . The column space of an $m \times n$ matrix is a subspace of $\mathbb{R}^m$.

Definition: Null Space

The null space of a matrix A , denoted $\text{Nul } A$, is the set of all solutions of the homogeneous equation $A\vec{x} = 0$.

Theorem 12

The null space of an $m \times n$ matrix A is a subspace of $\mathbb{R}^n$. Equivalently, the set of all solutions of a system $A\vec{x} = 0$ of m homogeneous linear equations in n unknowns is a subspace of $\mathbb{R}^n$.

Definition: Basis

A basis for a subspace H of $\mathbb{R}^n$ is a linearly independent set in H that spans H .

- The set $\{\vec{e}_1, \vec{e}_2, \dots, \vec{e}_n\}$ is the standard basis for $\mathbb{R}^n$.

2.9: Dimension and Rank

Definition: Coordinate Vector

Suppose the set $\mathcal{B} = \{\vec{b}_1, \dots, \vec{b}_p\}$ is a basis for a subspace H .

For each $\vec{x}$ in H , the **coordinates of $\vec{x}$ relative to the basis $\mathcal{B}$** are the weights $c_1, \dots, c_p$ such that:

$$\vec{x} = c_1\vec{b}_1 + \dots + c_p\vec{b}_p$$

The vector in $\mathbb{R}^p$:

$$[\vec{x}]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_p \end{bmatrix}$$

is called the **coordinate vector of $\vec{x}$ (relative to $\mathcal{B}$)** or the **$\mathcal{B}$ -coordinate vector of $\vec{x}$** .

Definition: Dimension

The **dimension** of a nonzero subspace H , denoted by $\dim H$, is the number of vectors in any basis for H .

The dimension of the zero subspace is defined to be 0.

Definition: Rank

The **rank** of a matrix, denoted by $\text{rank } A$, is the dimension of the column space of A (i.e., $\text{rank } A = \dim \text{Col } A$).

Theorem 14: The Rank Theorem

If matrix A has n columns, then:

$$\text{rank } A + \dim \text{Null } A = n$$

Theorem 15: The Basis Theorem

Let H be a p -dimension subspace of $\mathbb{R}^n$:

- Any linearly independent set of exactly p elements in H is automatically a basis for H .
- Any set of p elements of H that spans H is automatically a basis for H .

Theorem 8: The Invertible Matrix Theorem (Continued)

Let A be an $n \times n$ matrix. The following statements are each equivalent to " A is invertible":

- m. The columns of A form a basis of $\mathbb{R}^n$.
- n. $\text{Col } A = \mathbb{R}^n$.
- o. $\dim \text{Col } A = n$.
- p. $\text{rank } A = n$.
- q. $\text{Null } A = \{\vec{0}\}$.
- r. $\dim \text{Null } A = 0$.